Nikhil Rastogi

Bachelor of Technology
in Mathematics and Computing
Indian Institute Of Technology, Ropar

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GitHub | Portfolio
LinkedIn

Education

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	8.18 (Till 4th Sem)	2021-2025
Senior Secondary	Manav Rachna International School, Sec-46 (CBSE)	92.8%	2020
Secondary	Manav Rachna International School, Sec-46 (CBSE)	94.8%	2018

Experience

- **Google Summer of Code @ CHIPS Alliance** *May 2023 - Present*
  Google Open Source Remote
  – Added more front-end functionalities to the website
  – Shifting the codebase to ReactJs

Projects

- **Chat App** *Apr 2023 - Present*
  WebDev Project **Github**
  – MERN Stack Web App that allows users to chat in group and one-to-one
  – Uses Socket.io for real-time messaging, presence and online status, typing Indicators and more
- **Energy Trading dApp** *Jun 2023 - Present*
  Crewspace ICP team **Github**
  – dApp that enables peer-to-peer energy trading.
  – Built using Motoko (ICP) alongwith React.js
- **myNotes** *Jan 2023 - Feb 2023*
  WebDev Project **Github**
  – Web app for note taking which allows users to sign-in/sign-up and saves notes and edit them
  – Full-stack web application using the MERN stack
  – Employed RESTful API principles for smooth communication between the client-side and server-side
- **Encryption and Compression** *Oct 2022 - Dec 2022*
  DSA project **Github**
  – Made a project which accepts a string, encrypts it, compresses it, then decompresses it and then decrypts it.
  – Implemented RSA algorithm and Huffman compression in C++
- **Other Projects** *Dec 2021 - Dec 2022*
  WebDev Projects **Github**
  – Multiplayer Chess with videochat feature
  – Google Doc Clone - done using ReactJs, Socket.io, and MongoDB
  – Simple Todo List using ReactJs
  – Finance Tracker website which plots daily income and expenditure using MERN Stack
  – DailyBuzz - A web-based app showcasing daily news updates made using React.Js and NewsAPI
  – Text-Editor made using React.Js and Bootstrap
  – Leetcode Clone with ReactJs, Tailwind CSS, Firebase

Technical Skills

- **Programming Languages**: C/C++, Javascript, Python, Rust, Solidity, Motoko
- **Software/Tools**: MATLAB, Git/Github, VS Code, Command Prompt
- **Libraries/Frameworks**: Node.js, Bootstrap, JQuery, Express.js, React.js, EJS, Pug, Mongoose

Key courses taken

- **CSE & Maths**: Algorithm and Data Structure, Numerical Analysis using MATLAB, Theory of Computation, Discrete Mathematics, Real Analysis, Probability and Statistics
- **Others**: Tinkering Lab, Basic Electronics , Signals and Systems, Human Geography and Societal Needs, Economics

Positions of Responsibility

- **Secretary**, SoftCom Club, IIT Ropar *Jun. 2023 - Mar. 2024*
- **Coordinator**, ENarrators (Oratory) Club, IIT Ropar *Dec. 2022 - May 2023*
- **Coordinator**, Utility Feedback Management Committee, IIT Ropar *Dec. 2022 - May 2023*
- **Representative**, Mess Management Committee, IIT Ropar *Apr. 2022 - Oct. 2022*

Miscellaneous

- **JEE Advanced**, Achieved CRL 2653 *2021*
- **JEE Mains**, Achieved CRL 4008 *2021*
- **National Mathematics Talent Contest (NMTC)**, Qualified a national level exam taken by 66,000+ students *2017*